



AI Infrastructure Architecture Evolution

2015-2026

GPU · CPU · NVLink · NIC · DPU
Switching · Photonics · Systems · Software · Inference

A generation-by-generation synthesis of conference disclosures and primary technical sources.
Evidence cutoff: 28 September 2026.
Complete English edition, paired with a full Simplified Chinese report.

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01 Executive synthesis

NVIDIA's decade-long transition is best understood as a change in the unit of architecture: from a PCIe accelerator, to an NVLink-connected server, to a coherent CPU-GPU module, and finally to a rack-scale compute and networking platform. Tensor Core arithmetic is one part of that transition. Memory capacity, data movement, synchronization, network congestion, power delivery, and software scheduling increasingly determine how much of the arithmetic a deployed system can use.

Five changes define the product history. Pascal made HBM and GPU-to-GPU links central to the datacenter GPU. Volta established programmable matrix acceleration. Ampere broadened numerical formats and introduced practical multi-instance partitioning. Hopper reorganized execution around asynchronous tensor computation and data movement while Grace supplied a coherent, high-bandwidth CPU memory tier. Blackwell crossed the reticle boundary and moved the primary scale-up domain into the rack. Rubin extends this system approach with a new CPU core, HBM4, faster fabrics, and a more explicit separation between GPU computation, infrastructure processing, and latency-oriented inference.

This report covers 2015 through 28 September 2026. It is organized by product generation, not conference chronology. Hot Chips supplies high-level architecture; ISSCC supplies selected implementation disclosures; ISCA, MICRO, HPCA and ASPLOS supply research mechanisms and context. Vendor specifications, tuning guides and partner availability announcements connect those disclosures to commercial products. Research proposals are not treated as proof of adoption. The conference index is a curated map of the most useful disclosures, not an exhaustive bibliography of every NVIDIA-affiliated paper or appearance.

The main architecture transitions

Era	Flagship	Architectural change	System implication
2015-2016	M40 → P100	HBM2; NVLink 1	Move beyond host-mediated PCIe data movement
2017-2019	V100 / DGX-2	Tensor Cores; NVSwitch	Matrix acceleration plus a switched GPU domain
2020-2021	A100	BF16 / TF32 / FP64 Tensor Cores; MIG	One architecture serves AI, HPC and isolated instances
2022-2024	H100 / H200 / GH200	FP8; TMA; clusters; coherent C2C	Hide movement and extend the memory hierarchy
2024-2025	B200 / B300 / GB200 / GB300	Two compute dies; FP4; NVL72	Package and rack become coordinated design units
2026	Vera Rubin / BlueField-4 / LPX	HBM4; Olympus; NVLink 6; inference specialization	Co-design across compute, fabric, storage and serving

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Availability is a product-level attribute. A production announcement, an OEM shipment, a cloud service reaching general availability, and an MLPerf preview submission are different milestones. Blackwell has direct cloud GA evidence in 2025. Rubin has production disclosures and September 2026 preview results; those do not, by themselves, establish broad cloud GA. The report therefore records announced capabilities and the deployment milestone separately. Roadmap products remain separate from shipping comparisons.

02 Reading the specifications correctly

All compute rates are peak theoretical rates unless explicitly identified as measured results. A multiply-add counts as two operations. Dense and structured-sparse rates are kept separate; a sparse headline is not a dense capability. FP16 is not BF16, INT8 TOPS are not floating-point TFLOPS, and emulated FP64 is not native FP64. The report uses decimal GB/s and TB/s for interface bandwidth. A full-duplex total is divided by two only when the source defines a symmetric aggregate bidirectional rate. Switch capacity, endpoint injection rate, bisection bandwidth and the sum of all links are different quantities.

SKU identity matters as much as architecture identity. PCIe and SXM GPUs can differ in enabled SMs, clocks, HBM speed, power and NVLink support. B200 HGX specifications are not automatically GB200 specifications. Blackwell Ultra's dense and sparse FP4 modes do not share a universal 2:1 ratio. The HGX page gives B200 and B300 different dense FP4 rates despite the same sparse headline. Similarly, the Rubin chip-level peak and the power-managed NVL72 configuration expose different HBM and NVLink bandwidths. Both are retained with their configuration labels instead of selecting the larger number everywhere.

n/d means the reviewed primary sources did not establish the value at the required product scope; it does not mean zero. n/a means the field does not apply. Historical shipping means commercial deployment occurred, not that every old SKU remains orderable. GA columns use the first commercial year where established; an explicit announcement or preview label is not a GA date. Table panels retain the complete family schema. Bibliography IDs are collected in the source map at the end so the technical narrative remains readable.

03 GPU specification atlas

Datacenter GPU generations: representative configurations · 1/4

Generation	Codename / architecture	GA / milestone	Status	Process	Dies and packaging
M40	GM200 / Maxwell	2015	Historical shipping	28 nm	1 die; GDDR board
P100 SXM2	GP100 / Pascal	2016	Historical shipping	16 nm	1 die + HBM interposer
V100 SXM2	GV100 / Volta	2017	Historical shipping	12 nm	1 die + HBM interposer
A100 SXM 80 GB	GA100 / Ampere	2020	Historical shipping	7 nm	1 die + HBM interposer
H100 SXM	GH100 / Hopper	2022	Shipping	4N	1 die + HBM interposer
H200 SXM	GH100 / Hopper	2024	Shipping	4N	Hopper + HBM3e
B200 / HGX basis	Blackwell	2024-2025	Shipping	4NP	2 compute dies; NV-HBI; HBM
B300 / HGX basis	Blackwell Ultra	2025	Shipping	4NP	2 compute dies; NV-HBI; HBM
Rubin / chip peak	Rubin	2026 preview; GA n/d	Production / preview	n/d	2 compute dies; NV-HBI; HBM4

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Datacenter GPU generations: representative configurations · 2/4

Generation	Compute units	Dense BF16	Dense FP8	Dense FP6 / FP4	FP64 vector / matrix
M40	24 SM / 3072 CUDA	n/a	n/a	n/a	n/d / n/a
P100 SXM2	56 SM / 3584 CUDA	n/a	n/a	n/a	5.3 TF / n/a
V100 SXM2	80 SM / 640 TC	n/a	n/a	n/a	7.8 TF / n/a
A100 SXM 80 GB	108 SM / 432 TC	312 TF	n/a	n/a	9.7 / 19.5 TF
H100 SXM	132 SM / 528 TC	989 TF	1.979 PF	n/a	34 / 67 TF
H200 SXM	132 SM / 528 TC	989 TF	1.979 PF	n/a	34 / 67 TF
B200 / HGX basis	SKU-dependent	2.25 PF	4.5 PF	4.5 / 9 PF	37 TF published FP64; split n/d
B300 / HGX basis	Up to 160 SM / 640 TC; SKU-dependent	2.25 PF	4.5 PF	4.5 / 13.5 PF	1.25 TF published FP64; split n/d
Rubin / chip peak	224 SM / 896 TC	4 PF	17.5 PF training	17.5 / 35 PF training	33 TF native; 200 TF DGEMM emulated

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Datacenter GPU generations: representative configurations · 3/4

Generation	HBM: generation; stacks; GB; TB/s	Last-level cache	Scale-up: links; GB/s one-way; domain
M40	n/a; GDDR5 24 GB; 0.288 TB/s	3 MB	n/a
P100 SXM2	HBM2; 4; 16; 0.732	4 MB	4; 80; server topology
V100 SXM2	HBM2; 4; 16/32; 0.900	6 MB	6; 150; up to 16 via NVSwitch
A100 SXM 80 GB	HBM2e; 5 active; 80; 2.039	40 MB	12; 300; 8-GPU HGX
H100 SXM	HBM3; 5 active; 80; 3.35	50 MB	18; 450; 8-GPU HGX
H200 SXM	HBM3e; n/d; 141; 4.8	50 MB	18; 450; 8-GPU HGX
B200 / HGX basis	HBM3e; 8; 180; 8	126 MB GB200 reference; SKU-specific	18; 900; 8-GPU HGX
B300 / HGX basis	HBM3e; 8; up to 288; 8	n/d	18; 900; 8-GPU HGX
Rubin / chip peak	HBM4; 8; 288; 22	n/d	n/d; 1800 peak; NVL72

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Datacenter GPU generations: representative configurations · 4/4

Generation	Host link	TBP and cooling	Form factor
M40	PCIe 3 x16	250 W; air	PCIe
P100 SXM2	PCIe 3	300 W; platform-dependent	SXM2
V100 SXM2	PCIe 3	300 W; platform-dependent	SXM2
A100 SXM 80 GB	PCIe 4	400 W; platform-dependent	SXM4
H100 SXM	PCIe 5	700 W; platform-dependent	SXM5
H200 SXM	PCIe 5	700 W; platform-dependent	SXM5
B200 / HGX basis	PCIe 6	Up to 1000 W module; cooling by system	SXM
B300 / HGX basis	PCIe 6	n/d at uniform module basis	SXM
Rubin / chip peak	PCIe 6	Configuration-dependent; liquid	NVL / HGX

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

The atlas selects one reference configuration rather than merging every variant. P100 and V100 accelerate FP16 but do not have native BF16 Tensor Cores. A100’s original 40 GB version has 1.555 TB/s HBM bandwidth; the table uses the later 80 GB SXM version. B200/B300 compute figures are normalized from the HGX eight-GPU aggregate. B300 nominal memory is up to 288 GB per GPU in the DGX guide, while the HGX page lists 2.1 TB per eight-GPU platform; these are not silently equated. Product-specific memory accounting must be checked before capacity planning. The Rubin row is a preliminary chip-peak reference, not a promise that a rack sustains those rates simultaneously.

04 Maxwell and Pascal: establishing the accelerator system

Tesla M40 is the useful 2015 baseline: a graphics-derived Maxwell GPU adapted to datacenter deep learning, with GDDR5, a PCIe host connection and conventional CUDA arithmetic. Its value was parallel FP32 throughput and a mature programming platform, not dedicated tensor hardware. This establishes why a chronology beginning with Tensor Cores misses a major transition: before optimizing matrix instructions, NVIDIA had to improve the bandwidth and topology around the processor.

P100's GP100 die combines a compute-oriented SM, stronger FP64 capability, packed FP16 arithmetic, HBM2 and first-generation NVLink. Four HBM stacks move memory onto the advanced package; the resulting bandwidth comes with packaging complexity, stack capacity limits and thermal coupling. Four NVLink connections provide 80 GB/s in each direction in aggregate. That is a peer interconnect budget, not additional HBM bandwidth. PCIe still connects the host, and the actual usefulness of NVLink depends on which GPU pairs the server wires together.

Pascal also demonstrates why architecture names are insufficient for product comparisons. P4 and P40 use GDDR memory and emphasize efficient inference, including INT8, rather than reproducing GP100's HBM and HPC balance. P4's low-profile, low-power deployment target and P40's larger inference card are separate points on the same market map. A system architect should therefore track training/HPC and inference/graphics derivatives as branches, not treat every Pascal product as a smaller P100. Hot Chips 2016 explains the P100/NVLink system direction; product briefs distinguish the commercial branches.

05 Volta: matrix execution and switched scale-up

Volta's defining change is the Tensor Core: a matrix multiply-accumulate datapath exposed through CUDA programming abstractions and libraries. V100's peak FP16 tensor rate is not a replacement for its conventional FP32 or FP64 rate; it applies to a particular operand format, accumulation behavior and matrix execution path. Algorithms must produce suitable tiles and feed those units. The SM also separates integer and floating-point execution resources and introduces independent thread scheduling, changing the assumptions under which warp-synchronous code is safe.

Second-generation NVLink raises V100's aggregate one-way peer bandwidth to 150 GB/s. NVSwitch subsequently changes the server from a collection of selected direct links into a switched GPU fabric, exemplified by the 16-GPU DGX-2 domain. The important gain is not only link speed: a more regular topology makes all-to-all communication and collective scheduling easier to exploit. It does not remove collective algorithm costs, synchronization or memory placement decisions. HBM remains physically local to each GPU even when remote data is addressable.

Turing's T4 is the parallel inference branch. It combines Tensor Cores, low-precision inference support, video capability and a 70 W PCIe form factor with GDDR6 rather than HBM. That combination favors broad server deployment and inference density, not a large NVLink training domain. The distinction persists through L4, L40S and RTX PRO Blackwell: a shared numerical or shader lineage does not imply equivalent memory bandwidth, FP64 capability or scale-up connectivity.

06 Ampere: numerical breadth, isolation and implementation

A100 makes the tensor path useful to a wider range of workloads. BF16 preserves a large exponent range for training; TF32 gives FP32-facing workloads a tensor execution option with reduced mantissa precision; FP64 Tensor Cores extend matrix acceleration into HPC. Structured sparsity increases throughput only when the required sparsity pattern and software path are satisfied. This is an algorithmic contract, not a universal doubling of the chip. The 40 MB L2 cache and software-managed data movement help reuse operands before going back to HBM.

Multi-Instance GPU partitions A100 into as many as seven hardware-isolated GPU instances, with assigned compute and memory-system resources. This is materially different from merely time-sharing a process context. It enables smaller services to consume fractions of an expensive accelerator with stronger performance isolation. The trade-off is fragmentation: fixed profiles and memory capacity can strand resources when requests do not match the available partitions. Cluster placement, model size and scheduling remain necessary complements to the hardware.

The ISSCC 2021 A100 disclosure supplies a rare direct bridge from the marketed architecture to implementation: a 7 nm, approximately 54-billion-transistor, 826 mm² datacenter GPU. Combined with the Ampere whitepaper and Hot Chips material, it connects arithmetic, partitioning and I/O to a very large monolithic die. The 80 GB HBM2e refresh increases both capacity and bandwidth without a new SM generation. This is an early example of a recurring pattern: a memory refresh can materially improve model fit and utilization even when headline compute changes little.

07 Hopper and H200: asynchronous computation and memory

Hopper addresses the work required to keep tensor arithmetic busy. The Tensor Memory Accelerator moves multidimensional tiles with less thread involvement. Warp-group matrix instructions coordinate a larger execution group than the earlier warp-level path. Thread-block clusters and distributed shared memory allow cooperating blocks to share data and synchronize within a defined locality domain. Together these mechanisms encourage pipelined kernels: one stage prepares data while another computes. They do not eliminate resource constraints; registers, shared memory, barriers and cluster occupancy still bound the useful pipeline depth.

The Transformer Engine couples FP8 hardware with software-managed scaling and precision selection. E4M3 and E5M2 serve different numerical needs, so a throughput figure without format and accuracy conditions is incomplete. Hopper also adds DPX instructions for dynamic programming and strengthens confidential-computing support. Those capabilities widen the GPU's workload and deployment envelope; they should not be reduced to an LLM-only narrative. H100's fourth-generation NVLink supplies 900 GB/s aggregate bidirectional bandwidth, or 450 GB/s per direction, while an eight-GPU HGX fabric remains a distinct resource from its external network.

H200 retains Hopper computation and replaces the memory balance: 141 GB HBM3e at 4.8 TB/s, compared with H100 SXM's 80 GB at 3.35 TB/s. Capacity rises by about 1.76× and bandwidth by about 1.43×, while the selected dense BF16 peak is unchanged. This is particularly useful when inference is limited by weights or KV-cache traffic rather than arithmetic. It also changes how many devices a model needs simply to fit, potentially reducing communication. GH200 is a different product construction: a Grace CPU coherently attached to a Hopper GPU, with its own HBM configurations. H200's 141 GB must not be copied onto every GH200.

08 Blackwell and Ultra: crossing die and rack boundaries

Blackwell combines two large compute dies through a 10 TB/s NV-HBI connection and presents them as one CUDA GPU. That boundary is fundamentally different from connecting two separate GPUs with NVLink. The on-package fabric must support the internal behavior of one processor, including the interaction between compute and the distributed memory hierarchy. The approximately 208-billion-transistor design uses TSMC 4NP. Packaging makes the larger logical processor possible, but does not make die-to-die traffic free: locality, buffering, power and physical routing remain architectural costs.

Fifth-generation Tensor Cores extend low-precision execution to FP4 and FP6, with fine-grained scaling. NVFP4 uses block scaling plus a tensor-level scale to improve the usable numerical range of four-bit values. Tensor Memory supplies storage associated with tensor execution, reducing pressure on the conventional register path; two-SM cooperation changes how matrix work is distributed. These features require compiler, kernel and quantization support. A model's quality and latency target, not just supported bit width, determines whether FP4 is the right operating point.

GB200 places Grace alongside two Blackwell GPUs, and NVL72 connects 72 GPU packages into a rack-scale NVLink domain. GB300 upgrades that construction with Blackwell Ultra. The rack is not simply nine unrelated eight-GPU servers: communication topology, cooling, serviceability, power and scheduling are co-designed around a larger domain. Conversely, an NVL72 does not imply a uniform-latency 72-GPU memory pool. Software still decides tensor, pipeline, expert and data parallel placement, and crossing the rack boundary invokes a different fabric and congestion regime.

Ultra increases memory capacity and dense FP4 capability, but it is not a uniform multiplier across every datapath. The HGX comparison gives B300 much less native FP64 throughput than B200. That matters for mixed AI/HPC procurement and for numerical algorithms that cannot replace native FP64 with emulation. Cloud adoption also demonstrates configuration diversity: AWS P6e GB200 combines the GPU scale-up domain with EFA, while other deployments use NVIDIA networking. The GPU vendor's reference platform is therefore an important baseline, not a universal description of every hyperscaler installation.

09 Rubin: a platform generation, with configuration-specific limits

Rubin couples a new GPU generation to Vera, NVLink 6, ConnectX-9, BlueField-4 and Spectrum networking. The GPU moves to HBM4 and a larger tensor-compute budget; Vera changes the host CPU microarchitecture rather than merely raising Grace's clock. This increases the importance of balanced supply: more tensor work needs more operand bandwidth, collective bandwidth and orchestration capacity. The platform also introduces explicit support for infrastructure-side context storage and complementary low-latency inference, rather than requiring every task to occupy the same kind of GPU resource.

Two published Rubin configurations must remain distinct. The chip-oriented HGX specification lists 288 GB HBM4 at 22 TB/s and 3.6 TB/s aggregate bidirectional NVLink. The NVL72 page, under its DSX/MaxLPS at-scale configuration, lists 19.2 TB/s HBM bandwidth and 3 TB/s bidirectional NVLink per GPU. It also reports 0.45 TB/s bidirectional scale-out bandwidth per GPU. These values are operating-point and system-scope disclosures, not interchangeable descriptions of one unconstrained chip. Power management can trade instantaneous peaks for higher useful throughput across an entire facility.

The arithmetic table also needs execution-mode labels. Published Rubin training figures include dense FP4 at 35 PFLOPS and dense FP8/FP6 at 17.5 PFLOPS; the 50 PFLOPS NVFP4 inference headline has a sparse basis. It is unsafe to derive a single dense number by halving every headline. Native FP64 at 33 TFLOPS and an emulated FP64 DGEMM rate of 200 TFLOPS describe different mechanisms. An HPC application must validate accuracy, conditioning and library behavior before using the emulated rate in a performance model.

Hot Chips 2026 and the associated technical disclosures deepen the public architecture record. September 2026 MLPerf results are preview submissions, including a partner submission, and demonstrate working systems rather than a universal GA date. Rubin CPX remains a separately announced long-context branch in the evidence reviewed here: its original 2025 disclosure described GDDR7 and a late-2026 target, but this report does not infer final configuration or cancellation from later omission. Rubin Ultra and Feynman belong in the roadmap, not in shipping specification comparisons.

10 Grace and Vera CPUs: the host becomes architectural

CPU generations: one CPU chip unless specified · 1/4

Generation	Codename	GA / milestone	Status	Core µarch	Max cores / threads
Grace	Grace	2023	Shipping	Arm Neoverse V2	72 / 72
Vera	Olympus	2026 production / preview	GA n/d	NVIDIA Olympus	88 / 176

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

CPU generations: one CPU chip unless specified · 2/4

Generation	Process per die	Chiplets and packaging	L2 per core	L3 total / domain
Grace	4N	CPU + LPDDR; dual CPU or CPU-GPU C2C	1 MB	114 MB / chip
Vera	n/d	CPU + SOCAMM2; C2C platform	2 MB	164 MB / chip

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

CPU generations: one CPU chip unless specified · 3/4

Generation	Memory: type / channels / speed / GB/s	Sockets and socket links	PCIe / CXL
Grace	LPDDR5X; up to 480 GB; 512 GB/s; channel/speed SKU-dependent	1 or 2; C2C 900 GB/s bidirectional	64 lanes PCIe 5; CXL n/d
Vera	LPDDR5X SOCAMM2; up to 1.5 TB; 1200 GB/s; channel/speed n/d	1 or 2; C2C 1800 GB/s bidirectional	PCIe 6 / CXL 3.1; lane count n/d

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

CPU generations: one CPU chip unless specified · 4/4

Generation	Max TDP	Matrix / vector ISA
Grace	500 W dual-CPU Superchip including memory; single-chip n/d	Armv9.0-A; 4 × 128-bit SVE2
Vera	n/d	Armv9.2; 6 × 128-bit SVE2; FP8

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Grace's value is the CPU memory system and its connection to the accelerator. Seventy-two Neoverse V2 cores share a mesh and a large system cache, while LPDDR5X supplies high bandwidth with a different power and serviceability trade-off from conventional DIMMs. A two-CPU Grace Superchip doubles the cores and memory resources; GH200 instead couples Grace to Hopper. Coherent C2C allows fine-grained CPU-GPU sharing and access to a larger memory capacity without treating every transfer as an explicit bulk copy. Remote CPU memory remains slower for the GPU than local HBM, so coherence simplifies programming without erasing locality.

ISSCC 2023's NVLink-C2C disclosure makes this connection unusually concrete. It describes a 40 Gb/s-per-pin, single-ended off-package coherent interface, including a 5 nm PHY and clocking/skew techniques. Ten links support the disclosed 900 GB/s bidirectional interface. The older ISSCC 2018 ground-referenced-signaling research is relevant circuit lineage, but its 25 Gb/s-per-pin test vehicle is not the production Grace interface. This is precisely where conference synthesis is valuable: Hot Chips explains the memory-system purpose, while ISSCC reveals how an electrically demanding package boundary is implemented.

Vera replaces licensed Neoverse cores with NVIDIA's Olympus design. Its 88 cores expose 176 spatial threads, with a ten-wide front end, advanced prediction, memory renaming, value prediction and graph-oriented prefetching among the disclosed mechanisms. The architecture is aimed at host-side work that cannot simply be pushed into a large matrix operation: orchestration, irregular access, graph traversal and agent execution. Public descriptions do not establish every queue size or execution latency, so those are not reverse-engineered from marketing comparisons.

Vera's memory changes are equally consequential: up to 1.5 TB of replaceable SOCAMM2 LPDDR5X at 1.2 TB/s, larger private and shared caches, and twice Grace's C2C bidirectional rate. Using peak specifications, memory bandwidth per core rises from roughly 7.1 to 13.6 GB/s. Spatial threading does not double the memory system, so per-thread bandwidth is a different ratio. CXL 3.1 and Arm confidential-computing mechanisms broaden deployment options. The key trajectory is from a high-bandwidth accelerator companion to a CPU designed as a first-class component of the AI serving system.

11 Inference, workstation and edge branches

Related product branches

Product	Era	Memory	Deployment distinction
P4 / P40	2016	GDDR5	Pascal INT8 inference; not GP100 HBM/NVLink equivalence
T4	2018	16 GB GDDR6; 320 GB/s	70 W PCIe; inference and video; no NVLink
L4	2023	24 GB GDDR6; 300 GB/s	72 W Ada inference/video; no NVLink
L40S	2023	48 GB GDDR6; 864 GB/s	350 W; graphics plus AI; no NVLink
RTX PRO 6000 Blackwell Server Edition	2025	96 GB GDDR7; 1597 GB/s	Up to 600 W; graphics, AI and partitioned services
GB10 / DGX Spark	2025-2026	128 GB LPDDR5X; 273 GB/s	20 Arm cores; coherent shared memory; desktop AI; 140 W SoC TDP
Jetson AGX Thor	2025-2026	Edge configuration; not server HBM equivalence	Blackwell-derived edge/robotics branch; benchmark scope differs

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

These branches share programming tools and selected architectural mechanisms with datacenter accelerators, but optimize different constraints. GDDR-based cards can fit existing air-cooled servers and serve graphics, media and inference together. HBM-based GPUs prioritize bandwidth and large-scale compute fabrics. GB10 instead uses a coherent unified LPDDR memory system: its 128 GB capacity can be attractive for local model development, but 273 GB/s is far below a flagship HBM GPU’s bandwidth. Its one-PFLOPS FP4 headline should not be read as a one-PFLOPS dense training guarantee. The 240 W power supply rating is also not the 140 W SoC TDP.

12 NVLink, NVSwitch and coherent C2C

Interconnect generations · 1/2

Name / version	Year	Lane rate	Lanes per link	GB/s per link per direction	Links per device
NVLink 1	2016	20 Gb/s	8	20	4
NVLink 2	2017-2018	25 Gb/s	8	25	6
NVLink 3	2020	50 Gb/s	4	25	12
NVLink 4	2022	100 Gb/s	2	25	18
NVLink 5	2024-2025	n/d	n/d	50	18
NVLink 6	2026	n/d	n/d	n/d	n/d
NVLink-C2C / Grace	2023	40 Gb/s/pin	n/d	n/d	10
NVLink-C2C / Vera	2026	n/d	n/d	n/d	n/d

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Interconnect generations - 2/2

Name / version	Topology	Max / reference domain	Coherence / memory semantics
NVLink 1	Direct peer links	Server wiring-dependent	GPU peer access; not uniform memory
NVLink 2	Direct + NVSwitch	16 GPUs (DGX-2)	Peer memory; platform-specific CPU coherence
NVLink 3	NVSwitch	8 GPUs (HGX)	GPU peer memory
NVLink 4	NVSwitch	8 HGX; larger external switch systems	GPU peer memory / collectives
NVLink 5	NVSwitch	72 GPUs (NVL72)	Rack peer domain; not uniform latency
NVLink 6	NVSwitch	72 GPUs (NVL72)	3600 GB/s bidirectional peak per GPU; configured 3000
NVLink-C2C / Grace	CPU-CPU / CPU-GPU	Coherent module	900 GB/s bidirectional; coherent access
NVLink-C2C / Vera	CPU-GPU / CPU-CPU	Coherent platform	1800 GB/s bidirectional

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

NVLink Fusion opens selected NVLink system integration paths to custom CPUs and accelerators. It should be understood as an integration and ecosystem program, not as proof that every participating chip implements the same coherence semantics as Grace-Hopper. A custom processor must still satisfy interface, switch, software and platform requirements. This matters strategically because the scale-up fabric can remain NVIDIA-centered even when one compute component is supplied by another vendor. It does not mean Ethernet or InfiniBand disappear; those still serve scale-out and infrastructure roles.

13 ConnectX and BlueField: endpoints and infrastructure

NVIDIA's 2020 acquisition of Mellanox brought a mature network endpoint, switch and software portfolio into the compute platform. Three roles should remain separate. A ConnectX adapter is an efficient network endpoint with transport and offload engines. A SuperNIC is an AI-networking product role focused on the GPU back-end fabric; it does not automatically mean a general-purpose host CPU replacement. A BlueField DPU adds independently managed CPU and infrastructure processing for networking, storage and security. NVLink remains the separate scale-up fabric. Conflating these roles hides both trust boundaries and bandwidth bottlenecks.

NIC and DPU evolution - 1/4

Product	GA / milestone	Status	Class	Ports × speed
ConnectX-4 Lx	2015-2016 era	Historical shipping	NIC	25/50 GbE variants
ConnectX-5 / Ex	2017 era	Historical shipping	NIC	up to 2 × 100 Gb/s
ConnectX-6	2019 era	Historical shipping	NIC	up to 200 Gb/s
ConnectX-7	2022 era	Shipping	NIC / SuperNIC	1 × 400 / 2 × 200 Gb/s
ConnectX-8	2025	Shipping platform generation	SuperNIC	800 Gb/s class
ConnectX-9	2026	Production; configuration-dependent	SuperNIC	800 Gb/s card; up to 1600 Gb/s per GPU platform
BlueField	Pre-2020; exact GA n/d	Historical shipping	DPU	up to 2 × 100 Gb/s
BlueField-2	2020-2021 generation	Shipping	DPU	up to 200 Gb/s
BlueField-3	2023 generation	Shipping	DPU / SuperNIC	up to 400 Gb/s
BlueField-4	2026	Production disclosure	DPU	800 Gb/s

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

NIC and DPU evolution - 2/4

Product	SerDes	Host interface	Programmable engine
ConnectX-4 Lx	25G NRZ	PCIe 3 x8/x16	eSwitch / steering
ConnectX-5 / Ex	25G NRZ	PCIe 3 x16; Ex PCIe 4	eSwitch / steering
ConnectX-6	50G PAM4	PCIe 4 x16	Transport / steering
ConnectX-7	100G PAM4	PCIe 5 x16; Socket Direct	Transport / steering
ConnectX-8	n/d	PCIe 6 + integrated switch	Programmable transport / steering
ConnectX-9	200G class	PCIe 6 x16 / Socket Direct x32	AI transport / steering
BlueField	25G class	up to 32 lanes PCIe 3/4	ConnectX-5 eSwitch
BlueField-2	50G class	PCIe 4 x16	ConnectX-6 Dx + Arm
BlueField-3	100G class	up to 32 lanes PCIe 5	ConnectX-7 + DPA + Arm
BlueField-4	200G class	PCIe 6	ConnectX-9 + Grace

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

NIC and DPU evolution - 3/4

Product	Embedded CPU	On-card memory	Transport / RDMA
ConnectX-4 Lx	n/a	Adapter buffers; capacity n/d	Ethernet / RoCE
ConnectX-5 / Ex	n/a	n/d	Ethernet / IB / RoCE
ConnectX-6	n/a	n/d	Ethernet / HDR IB / RoCE
ConnectX-7	n/a	n/d	Ethernet / NDR IB / RoCE
ConnectX-8	n/a	n/d	Ethernet / IB / RoCE
ConnectX-9	n/a	n/d	Ethernet / XDR IB / RoCE
BlueField	up to 16 Arm A72	DDR4; capacity SKU-dependent	Ethernet / IB / RoCE
BlueField-2	8 Arm A72	DDR4; SKU-dependent	Ethernet / RoCE; variants
BlueField-3	up to 16 Arm A78	16/32 GB DDR5; SKU-dependent	Ethernet / NDR IB / RoCE
BlueField-4	64 Neoverse V2	LPDDR5X; 250 GB/s	Ethernet / RDMA

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

NIC and DPU evolution - 4/4

Product	Congestion control	Key offloads	Power
ConnectX-4 Lx	ECN / PFC	Overlay, virtualization, RDMA	SKU-dependent
ConnectX-5 / Ex	ECN / PFC	GPUDirect RDMA; storage; switching	SKU-dependent
ConnectX-6	ECN / PFC	GPUDirect RDMA; variant-specific crypto	SKU-dependent
ConnectX-7	Adaptive routing / ECN by fabric	GPUDirect; virtualization; crypto	SKU-dependent
ConnectX-8	AI adaptive routing / congestion control	GPU/NIC PCIe switching; GPUDirect	SKU-dependent
ConnectX-9	Spectrum-X multiplane / adaptive routing	GPUDirect; PCIe fabric; security	SKU-dependent
BlueField	ECN / PFC	NVMe-oF; switching; virtualization	SKU-dependent
BlueField-2	ECN / PFC	DOCA; storage; crypto; virtual switch	SKU-dependent
BlueField-3	Spectrum-X / adaptive congestion control	DOCA; storage/security; asynchronous offload	SKU-dependent
BlueField-4	Platform-dependent	Infrastructure OS; STX / context storage; security	n/d

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

The ConnectX trajectory follows both network SerDes and the host interface. Moving from 100 to 200, 400 and 800 Gb/s requires sufficient PCIe bandwidth and suitable server wiring. Socket Direct can split connectivity across two CPU roots; an advertised aggregate port rate does not guarantee that a single older PCIe link can sustain it. ConnectX-8's integrated PCIe Gen6 switch is therefore a system feature, not just a faster MAC: it changes GPU-to-NIC attachment and permits high-bandwidth local paths without assuming every CPU root complex has the newest PCIe generation.

BlueField evolves from an Arm-plus-ConnectX storage/network processor into a separately managed infrastructure subsystem. BlueField-3 adds a programmable data-path accelerator alongside Arm control cores and fixed-function packet engines. The Arm cores should not be assumed to forward every packet in software at line rate. BlueField-4 combines Grace-class CPU resources with ConnectX-9 and targets infrastructure services, including storage for AI context. Moving KV-cache or context data outside GPU HBM can improve capacity economics, but introduces a hierarchy: storage/network bandwidth and access granularity must match the serving scheduler's decisions.

14 Spectrum, Quantum and photonics

Switch and fabric evolution

Family	Milestone	Published capacity	Architecture / deployment meaning
Spectrum-1	2015 generation	3.2 Tb/s	Ethernet switching baseline
Spectrum-2	2018-2019 generation	6.4 Tb/s	50G-class lanes; 200GbE ports
Spectrum-3	2020 generation	12.8 Tb/s	400GbE generation
Spectrum-4 / Spectrum-X	2022 chip; 2023 platform	51.2 Tb/s	400/800GbE; switch + SuperNIC + software co-design
Spectrum-6	2026 production / adoption	102.4 Tb/s ASIC	Multiplane AI Ethernet; Vera Rubin ecosystem
Spectrum-X / Quantum-X photonics	2025-2026 disclosures	Up to 409.6 Tb/s Spectrum-X assembly; not one ASIC	Co-packaged optics; product configuration and rollout vary
Quantum-X800 / Q3450-LD	2025-2026 disclosures	144 × 800 Gb/s	CPO-based InfiniBand switch; 200G SerDes; liquid cooling
Quantum-2	2021 announcement; 2022 generation	64 × 400 Gb/s = 25.6 Tb/s one-way	NDR InfiniBand; adaptive routing; SHARP collectives

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Spectrum-X is a coordinated Ethernet platform rather than a switch ASIC renamed for AI. Its value comes from the interaction of adaptive packet routing, endpoint congestion response, telemetry, ordering support and collective software. Multiplane designs distribute traffic across parallel network planes and use endpoint intelligence to avoid hotspots. These mechanisms address the synchronized bursts and all-to-all exchanges common in distributed training and mixture-of-experts inference. A nominally nonblocking topology can still perform poorly if congestion control reacts too slowly or tenant traffic interferes.

Quantum serves the InfiniBand branch, with a transport and management model designed for tightly coupled clusters. SHARP moves selected collective reduction work into the network; it does not execute arbitrary GPU kernels inside the switch. Comparing Quantum with Spectrum therefore requires matching the collective, topology, software and operational model, not only port speed. Quantum-2 illustrates a common accounting trap: 64 ports at 400 Gb/s are 25.6 Tb/s in one direction; a 51.2 Tb/s full-duplex total is not twice the one-way switching capacity.

Co-packaged optics shortens the electrical path between switch silicon and optical conversion. The expected system benefit is lower electrical I/O power and improved reach at very high aggregate bandwidth, but it changes packaging, laser supply, fiber attachment, repair and qualification requirements. A photonic switch assembly may contain multiple switching elements; its aggregate capacity must not be assigned to one ASIC. ISSCC optical-link research and product CPO disclosures belong in the same technology map, but a research test chip is not proof of the exact production optical implementation.

15 Groq 3 LPX and specialized inference

The 2025 agreement between Groq and NVIDIA is a non-exclusive inference-technology licensing agreement, not an acquisition of Groq as a company. Groq 3 LPX subsequently appears as a complementary NVIDIA inference platform, with production disclosed in August 2026. This changes the portfolio map: NVIDIA does not need a merchant FPGA line to have a non-GPU inference architecture. The correct comparison is between execution and memory organizations, not merely between two vendors' TOPS figures.

LPX organizes computation around compiler-scheduled execution and substantial on-chip SRAM. The disclosed LP30 LPU has approximately 500 MB SRAM and 150 TB/s local SRAM bandwidth; a 256-LPU rack aggregates 128 GB SRAM and about 40 PB/s of local bandwidth. Those sums are not a uniform 128 GB cache with 40 PB/s available to any one operator. The compiler must place and stream model state across devices and their interconnect. Small, predictable working sets can exploit low-latency local access; large models still confront distributed capacity and communication constraints.

The architectural attraction is heterogeneous serving. HBM GPUs offer large capacity and flexible computation; SRAM-centric LPUs offer a different latency/bandwidth operating point. A serving system can place suitable work on each, but the split must account for transfer costs, batching, KV-cache location and end-to-end latency. It is not enough for one feed-forward stage to become faster if the interface or another stage becomes the critical path. LPX therefore belongs beside the GPU and DPU chapters as a complementary execution tier, not inside a table pretending it is another CUDA GPU.

16 Integrated systems and the bandwidth hierarchy

System reference configurations · 1/2

Platform	Year	CPUs / accelerators	Scale-up domain
DGX-1 / P100	2016	x86 + 8 P100	8 GPUs; nonuniform direct topology
DGX-2 / V100	2018	x86 + 16 V100	16 GPUs / NVSwitch
HGX / DGX A100	2020	x86 + 8 A100	8 GPUs / NVSwitch
HGX H100 / H200	2022-2024	x86 + 8 GPUs	8 GPUs / NVSwitch
GH200 systems	2023-2024	Grace + Hopper	Coherent module; wider topology by system
GB200 NVL72	2025	36 Grace + 72 Blackwell	72 GPU packages
GB300 NVL72	2025	36 Grace + 72 Blackwell Ultra	72 GPU packages
Vera Rubin NVL72	2026	36 Vera + 72 Rubin	72 GPU packages
Groq 3 LPX	2026	256 LPUs; 32 trays	LPU fabric; not NVLink GPU domain

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

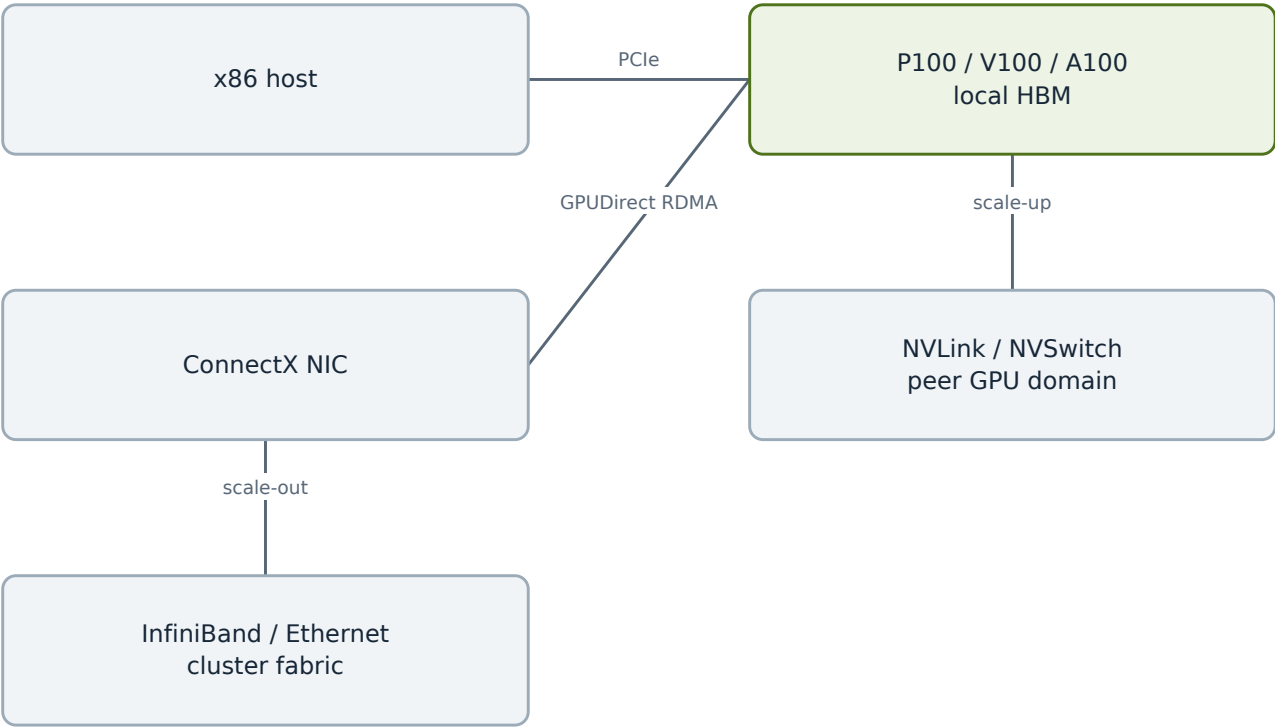
System reference configurations · 2/2

Platform	Scale-out per accelerator	Rack / system power	Cooling
DGX-1 / P100	configuration-dependent	n/d	Air
DGX-2 / V100	configuration-dependent	n/d	Air
HGX / DGX A100	configuration-dependent	n/d	Platform-dependent
HGX H100 / H200	configuration-dependent	n/d	Air / liquid by system
GH200 systems	configuration-dependent	n/d	Platform-dependent
GB200 NVL72	Reference / cloud-specific; AWS 400 Gb/s EFA per GPU	configuration-dependent	Liquid
GB300 NVL72	configuration-dependent	configuration-dependent	Liquid
Vera Rubin NVL72	200 GB/s one-way reference; 225 GB/s MaxLPS page	configuration-dependent	Liquid
Groq 3 LPX	n/d	n/d	Liquid

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

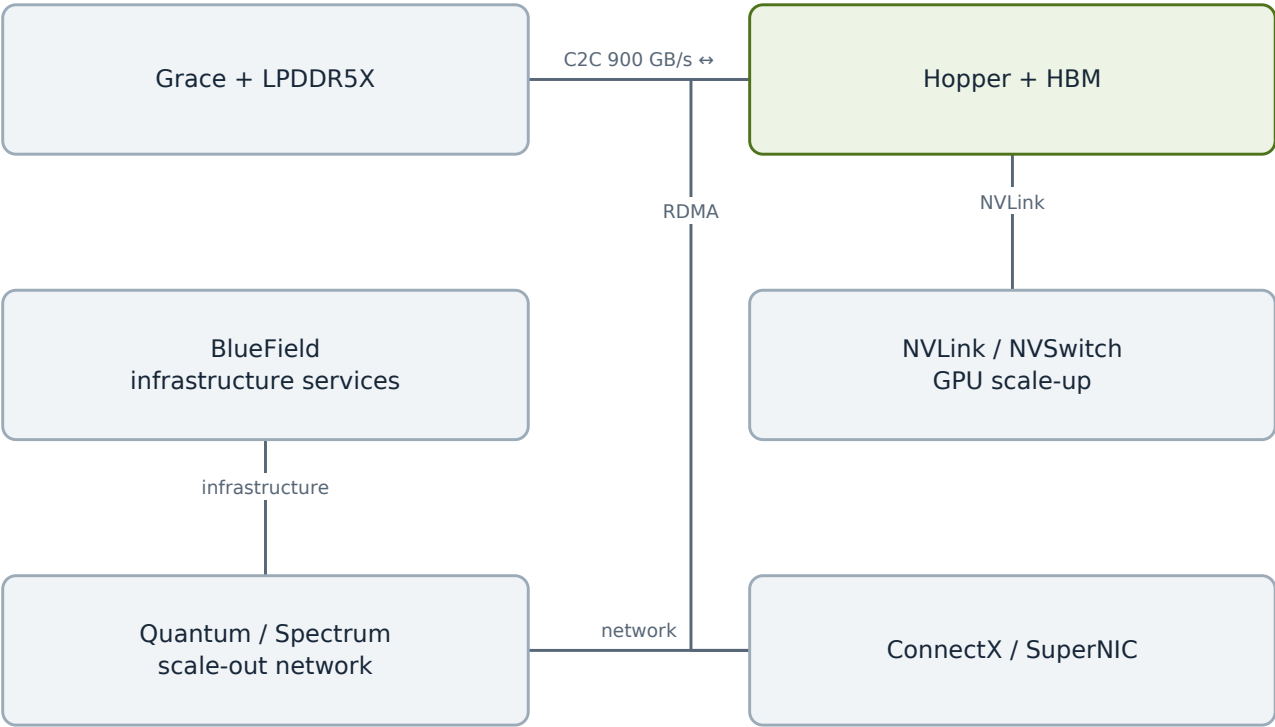
DGX is an integrated system product; HGX is a GPU baseboard/platform building block; MGX is a modular server and rack design ecosystem; NVL describes an NVLink-centered system configuration. These labels should not be used interchangeably. OEMs can vary CPU, network, storage, cooling and power around common GPU building blocks. Rack power also includes CPUs, switches, fans or pumps, memory and conversion losses, so multiplying GPU TDP by GPU count is not a rack-power specification. Power-limited operating points further break that shortcut.

2016-2020: the GPU server becomes a fabric



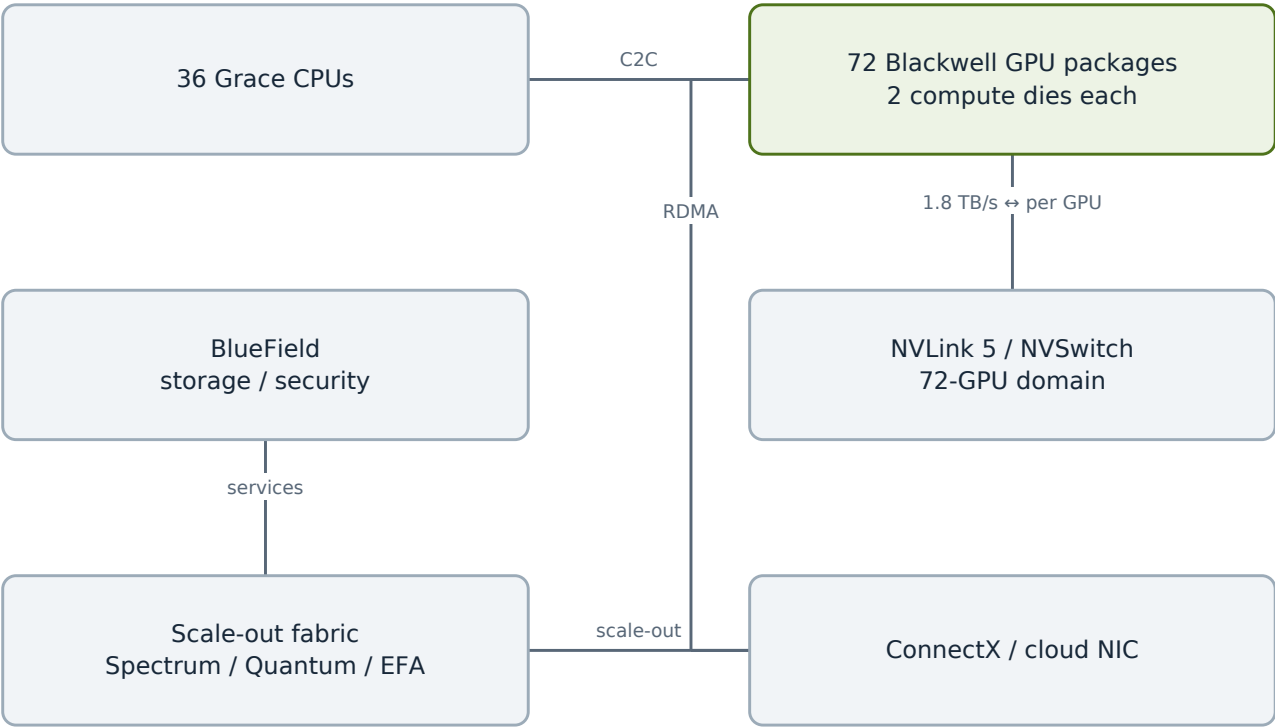
Logical connectivity, not a board wiring diagram. The GPU-local HBM and external network remain separate bandwidth domains.

2023-2024: coherent CPU-GPU memory



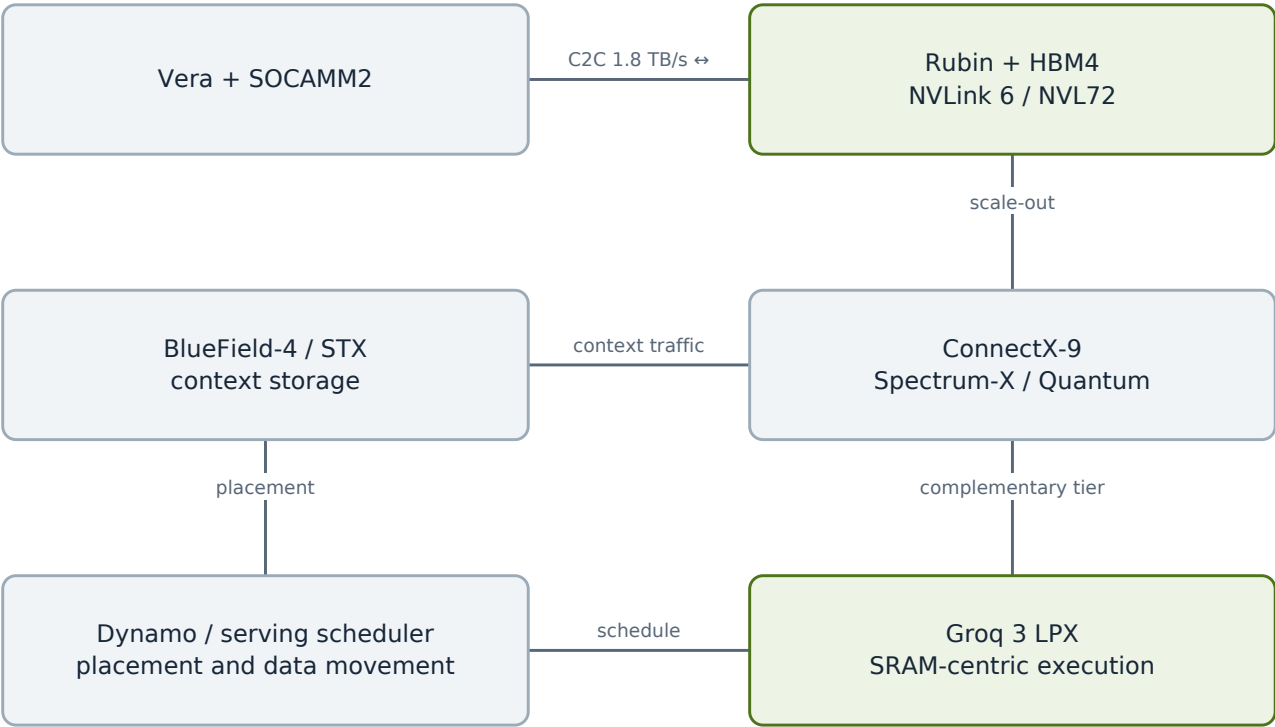
C2C enables coherence; it does not make LPDDR latency or bandwidth equal to local HBM. Arrows show connectivity, not traffic direction.

2025: the NVL72 rack as a scale-up unit



Cloud-specific networks are alternatives, not a claim that all three fabrics are installed together.

2026: heterogeneous serving and infrastructure



Conceptual placement map; the LPX edge does not assert a direct NVLink or coherent memory connection to Rubin.

Normalized comparisons and derived implications

Comparison	Calculation	Interpretation
H200 / H100 capacity	$141 / 80 = 1.76\times$	More model / KV capacity without a new SM generation
H200 / H100 HBM bandwidth	$4.8 / 3.35 = 1.43\times$	Bandwidth gain differs from capacity gain
Grace peak memory per core	$512 / 72 = 7.11 \text{ GB/s}$	Aggregate divided by cores; not a guarantee per core
Vera peak memory per core	$1200 / 88 = 13.64 \text{ GB/s}$	1.92× Grace on this basis
Rubin HBM / NVLink, MaxLPS	$19.2 / (3 / 2) = 12.8\times$	Local HBM bandwidth exceeds one-way peer injection
Rubin NVLink / scale-out, MaxLPS	$(3 / 2) / (0.45 / 2) = 6.67\times$	Rack-local placement can avoid a slower network tier
800 Gb/s network conversion	$800 / 8 = 100 \text{ GB/s}$	Line rate before protocol and PCIe overheads

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

These ratios expose the hierarchy without predicting application speed. A kernel’s roofline is the minimum of compute throughput and operational intensity times the relevant memory bandwidth. Distributed execution adds communication and synchronization ceilings. Moving from HBM to peer GPU memory, CPU memory, network-attached context storage or LPX SRAM changes both the bandwidth and the ownership of data. The architectural task is to place reuse at the cheapest sufficiently large tier, not to add the advertised bandwidths together.

17 Software makes the hardware hierarchy usable

Software and programming milestones

Release / family	Date / era	Hardware enabled	Key features
CUDA / Pascal	2016	P100	Unified memory evolution; Pascal programming
CUDA / Volta	2017	V100	Tensor programming; independent scheduling
Ampere software generation	2020	A100	TF32 / BF16 libraries; MIG; sparsity support
Hopper software generation	2022-2024	H100 / H200 / GH200	Transformer Engine; TMA; clusters; FP8
Blackwell software generation	2024-2026	B200 / B300 / GB200 / GB300	FP4 / FP6; Tensor Memory; revised kernels
NCCL / NVSHMEM ecosystem	Cross-generation	GPU fabrics	Collective communication and GPU-oriented communication; topology-aware algorithms
DOCA	BlueField era	BlueField / ConnectX	Infrastructure offload APIs and services
Dynamo	2025 onward	Distributed inference	Disaggregated serving; scheduling and data movement

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

CUDA provides the execution and memory model, while cuBLAS, cuDNN, CUTLASS and related libraries turn architectural features into optimized kernels. TensorRT and TensorRT-LLM specialize inference execution; NCCL coordinates collectives across the actual topology. These layers are not interchangeable. A Tensor Core format can exist in silicon before every framework or model uses it well. Likewise, a new network adapter does not automatically accelerate a collective unless the transport, topology discovery and algorithm can exploit it.

Dynamo reflects the shift from optimizing one model invocation to managing a distributed serving pipeline. Prefill, decode, expert placement and KV-cache movement have different resource profiles. Separating them can improve utilization, but also creates network traffic and scheduling dependencies. BlueField context storage and LPX introduce more placement choices, making the scheduler part of the architecture. Performance comparisons should therefore fix model, precision, quality target, input/output length, batching and latency objective before attributing gains to a new GPU generation.

18 What the architecture conferences actually establish

ISCA 2017's MCM-GPU explores how several GPU modules could behave as a scalable logical accelerator and why inter-module locality matters. MICRO 2017's Beyond the Socket studies NUMA-aware GPUs; Fine-Grained DRAM studies energy-efficient memory organization. HPCA 2015 and ASPLOS 2015 examine bandwidth and page placement in heterogeneous coherent-memory systems. These papers explain persistent design pressures visible later in Blackwell and Grace: packaging boundaries, nonuniform bandwidth, placement and locality. They do not establish that a later commercial product implements the paper's exact mechanism.

MICRO 2019's Simba is especially useful as a counterpoint to the general-purpose GPU: it studies a multi-chip inference accelerator with distributed local storage and a deliberately structured communication hierarchy. Its circuit implementation was disclosed at VLSI and in journal work; it should not be mislabeled an ISSCC 2019 GPU product paper. COPA-GPU, published in the ACM TACO line, explores composing different memory and compute organizations for different domains. Both broaden the design-space discussion without serving as a hidden specification for Blackwell or LPX.

Direct product linkage is strongest where the source explicitly names it. ISSCC's A100 and NVLink-C2C disclosures are implementation evidence for those products. NVIDIA's account of PrefixRL states that Hopper includes thousands of arithmetic circuits designed using the method; that supports a concrete design-tool adoption claim, not the statement that the whole GPU was designed by AI. Hot Chips 2026 names Vera, Rubin, BlueField-4 and LPU presentations, but a program entry proves that a talk occurred, not that every slide or undisclosed microarchitectural detail was publicly inspected.

19 Conference-to-product index

Selected disclosures and their evidentiary role · 1/2

Venue	Year	Paper / talk	Presenters / authors
Hot Chips	2016	Ultra-Performance Pascal GPU and NVLink Interconnect	Denis Foley
Hot Chips	2017	NVIDIA's Volta GPU: Programmability and Performance for GPU Computing	Jack Choquette
Hot Chips	2020	A100 / Ampere architecture	NVIDIA
Hot Chips	2022	NVIDIA's Hopper GPU: Scaling Performance	Jack Choquette
Hot Chips	2022	NVIDIA's Grace CPU	NVIDIA
Hot Chips	2022	NVLink-Network Switch	NVIDIA
Hot Chips	2024	NVIDIA Blackwell Platform: Advancing Generative AI and Accelerated Computing	Ajay Tirumala; Raymond Wong
Hot Chips	2025	GB200 NVL72 case study	John Norton
Hot Chips	2025	ConnectX-8 SuperNIC: A Programmable RoCE Architecture for AI Data Centers	Idan Burstein
Hot Chips	2025	Co-Packaged Silicon Photonics Switches for Gigawatt AI Factories	Gilad Shainer
Hot Chips	2025	NVIDIA's GB10 SoC: AI Supercomputer On Your Desk	Andi Skende
Hot Chips	2025	RTX 5090: Designed for the Age of Neural Rendering	Marc Blackstein

Venue	Year	Paper / talk	Presenters / authors
Hot Chips	2026	Vera CPU	Jonathon Evans; Polychronis Xekalakis
Hot Chips	2026	Rubin GPU: Driving the Era of Agentic AI	Manas Mandal; Rajballav Dash; Rouslan Dimitrov
Hot Chips	2026	BlueField-4 Processor Powers the AI Factory Operating System	Idan Burstein
Hot Chips	2026	Spectrum-X Multiplane Network Architecture	Gilad Shainer
Hot Chips	2026	Think Fast: LPU Accelerator for Heterogeneous Compute	Igor Arsovski; Santosh Raghavan
ISSCC	2018	1.17pJ/b 25Gb/s/pin Ground-Referenced Single-Ended Serial Link	NVIDIA research team
ISSCC	2021	3.2 The A100 Datacenter GPU and Ampere Architecture	Jack Choquette et al.
ISSCC	2023	9.3 NVLink-C2C: A Coherent Off-Package Chip-to-Chip Interconnect with 40Gbps/pin Single-Ended Signaling	Ying Wei et al.
ISSCC	2026	17.1 GB10 SoC Built for AI Acceleration	Andi Skende; G. Rosseel; N. Pinckney
ISCA	2017	MCM-GPU: Multi-Chip-Module GPUs for Continued Performance Scalability	NVIDIA research collaborators
MICRO	2017	Beyond the Socket: NUMA-Aware GPUs	Ugljesa Milic et al.
MICRO	2017	Fine-Grained DRAM: Energy-Efficient DRAM for Extreme Bandwidth Systems	Mike O'Connor et al.
MICRO	2019	Simba: Scaling Deep-Learning Inference with Multi-Chip-Module-Based Architecture	Yakun Sophia Shao et al.
HPCA	2015	Unlocking Bandwidth for GPUs in CC-NUMA Systems	Neha Agarwal et al.

Venue	Year	Paper / talk	Presenters / authors
ASPLOS	2015	Page Placement Strategies for GPUs within Heterogeneous Memory Systems	Neha Agarwal et al.
ACM TACO	2021	GPU Domain Specialization via Composable On-Package Architecture	NVIDIA research collaborators
DAC / follow-up	2021-2022	PrefixRL / arithmetic circuit design	NVIDIA research team
GTC	2016-2026	P100 → Volta → Ampere → Hopper → Blackwell → Rubin	NVIDIA
Computex	2025	ConnectX-8 and NVLink Fusion	NVIDIA
SC20	2020	A100 80 GB	NVIDIA
OCP Global Summit	2025	MGX, power delivery and AI-factory infrastructure	NVIDIA / partners
AWS re:Invent	2025	NVIDIA platforms and NVLink Fusion collaboration	AWS / NVIDIA
Microsoft Ignite	2025	Azure ND GB300 v6 GA	Microsoft
Google Cloud Next / update	2025	A4X / GB200	Google Cloud
MWC / AI-RAN	2025	Aerial AI-RAN architecture	NVIDIA / telecom partners
AI Infra Summit	2026	Vera Rubin / DSX efficiency	NVIDIA

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Selected disclosures and their evidentiary role · 2/2

Venue	Year	Product / generation	Disclosure type	Source / link ID
Hot Chips	2016	P100 / NVLink 1	Product architecture	C01
Hot Chips	2017	V100	Product architecture	C02
Hot Chips	2020	A100	Product architecture	C03
Hot Chips	2022	H100	Product architecture	C04
Hot Chips	2022	Grace	CPU / memory system	C04
Hot Chips	2022	NVLink switch	Scale-up architecture	C04
Hot Chips	2024	Blackwell / GB200	Product and system architecture	C05 C11
Hot Chips	2025	GB200 NVL72	System tutorial	C06
Hot Chips	2025	ConnectX-8	Network endpoint architecture	C06 P19
Hot Chips	2025	Photonics	Optical networking	C06 P23
Hot Chips	2025	GB10	SoC architecture	C06
Hot Chips	2025	Client Blackwell	Related graphics branch	C06
Hot Chips	2026	Vera	Program + vendor technical disclosure	C07 P10
Hot Chips	2026	Rubin	Program + vendor technical disclosure	C07 P07
Hot Chips	2026	BlueField-4	Program + vendor technical disclosure	C07 P20

Venue	Year	Product / generation	Disclosure type	Source / link ID
Hot Chips	2026	Spectrum-X	Program + vendor technical disclosure	C07 P39
Hot Chips	2026	LPX	Program + vendor technical disclosure	C07 P12
ISSCC	2018	Signaling research	Research circuit; not Grace production PHY	R07
ISSCC	2021	A100	Official press kit; direct implementation	C08
ISSCC	2023	Grace / Hopper C2C	Official press kit; PHY implementation	C09
ISSCC	2026	GB10	Official program; full paper not inspected	C10
ISCA	2017	Multi-chip GPU research	Research; no automatic Blackwell attribution	R01
MICRO	2017	NUMA GPU research	Research mechanism	R02
MICRO	2017	Memory research	Research mechanism	R03
MICRO	2019	Simba	Research accelerator	R10
HPCA	2015	Coherent-memory placement	Research; not Grace specification	R04
ASPLOS	2015	Page placement	Research; not Grace specification	R05
ACM TACO	2021	COPA-GPU	Research; journal, not a Big Four talk	R06
DAC / follow-up	2021-2022	Hopper design flow	Explicit product adoption in follow-up	P41

Venue	Year	Product / generation	Disclosure type	Source / link ID
GTC	2016-2026	Full platform	Launch, architecture, software, roadmap	P01 P02 P03 P04 E03 E07
Computex	2025	Network and ecosystem	Product / ecosystem	P19 P27
SC20	2020	A100 / HGX	Memory refresh announcement	E20 P15
OCP Global Summit	2025	Rack ecosystem; future power designs	System integration / roadmap	E12
AWS re:Invent	2025	Blackwell / Fusion	Ecosystem; GA verified separately	E13 E16
Microsoft Ignite	2025	GB300 NVL72	Cloud GA	E15
Google Cloud Next / update	2025	GB200	Cloud product; May GA update	E14
MWC / AI-RAN	2025	Grace Hopper / Blackwell / BlueField	Deployment architecture; not GPU core disclosure	E19
AI Infra Summit	2026	Rubin	System power and efficiency	E11

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

20 Cross-generation conclusions and open disclosures

The central trend is specialization inside a programmable system, not a simple replacement of programmability by fixed function. Tensor Cores specialize arithmetic; TMA specializes data movement; NVSwitch specializes scale-up communication; SHARP specializes selected reductions; BlueField specializes infrastructure; LPX specializes a different inference execution regime. CUDA and the serving stack bind these resources together. The system becomes more capable, but also more dependent on placement, numerical choices and coordinated software.

The durable engineering constraints are clear. HBM capacity and bandwidth determine model fit and reuse. Die-to-die interfaces let the processor grow but impose locality costs. Rack-scale fabrics reduce some communication penalties while raising power, cooling and serviceability demands. Faster network line rates require host-interface bandwidth and congestion control. Coherent CPU memory and external context stores expand capacity without becoming substitutes for local HBM performance. Each generation moves these boundaries; none abolishes them.

Important public gaps remain: complete CPU queue and pipeline parameters; some enabled-unit counts and power points by Blackwell SKU; reconciled usable-memory accounting across DGX and HGX pages; final Rubin configuration and broad GA milestones; production optical implementation details; and the final commercial configuration of CPX. These gaps are concentrated here rather than repeated as confidence labels throughout the report. They should be closed with dated primary disclosures, not by averaging inconsistent specifications or importing rumor into a product table.

21 Codename crosswalk and source guide

Architecture, silicon and product names

Architecture / silicon	Products	Important distinction
Maxwell / GM200	M40	No Tensor Cores
Pascal / GP100	P100	GP100 differs from P4/P40 inference dies
Volta / GV100	V100	Turing T4 is a separate branch
Ampere / GA100	A100	GA100 differs from graphics Ampere
Hopper / GH100	H100 / H200	GH200 names a Grace-Hopper combination
Datacenter Blackwell	B200 / GB200	B200 GPU vs GB200 CPU-GPU construction
Blackwell Ultra	B300 / GB300	Not the same FP64 balance as B200
Rubin + Vera	Vera Rubin NVL72	Rubin GPU and Vera CPU are distinct chips
Grace-derived + ConnectX-9	BlueField-4	DPU CPU is not Vera
LP30 / LPU	Groq 3 LPX	Licensed inference technology; not a CUDA GPU

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Sources are grouped below into product documentation, conference records, research papers and event/deployment disclosures. The source map associates each report section with its evidence. The accompanying canonical data file preserves table-cell claims, basis, product scope, source IDs and dates for future updates. Conference programs and press kits are identified as such; this report does not imply access to paywalled papers or restricted recordings. Specifications are a snapshot at the cutoff, and live product pages may subsequently change.

Source map

- 01 Executive synthesis - C01, C04, C08, C09, E01, E08, E09, E14, E15, E16, P01, P02, P03, P04, P05, P06, P07, P08, P10, P11, P12, P14, P37, P42, R01, R04, R05
- 02 Reading the specifications correctly - E08, E14, E15, E16, P01, P02, P03, P06, P11, P13, P15, P37, P38
- 03 GPU specification atlas - E08, E18, P01, P02, P03, P04, P05, P06, P07, P11, P13, P14, P15, P16, P17, P32, P33, P34, P35, P36, P37, P42
- 04 Maxwell and Pascal: establishing the accelerator system - C01, E18, P01, P36, P42
- 05 Volta: matrix execution and switched scale-up - C02, P02, P18, P28, P29, P30, P31, P35
- 06 Ampere: numerical breadth, isolation and implementation - C03, C08, P03, P15, P34
- 07 Hopper and H200: asynchronous computation and memory - C04, P04, P08, P13, P14, P18, P33
- 08 Blackwell and Ultra: crossing die and rack boundaries - C11, E14, E15, E16, P05, P06, P18, P27, P32, P37
- 09 Rubin: a platform generation, with configuration-specific limits - C07, E03, E04, E08, E11, P07, P10, P11, P12, P20, P37, P38
- 10 Grace and Vera CPUs: the host becomes architectural - C07, C09, E08, P08, P09, P10, P11, R07
- 11 Inference, workstation and edge branches - C10, E08, E18, P28, P29, P30, P31, P46
- 12 NVLink, NVSwitch and coherent C2C - C09, E13, P01, P02, P03, P04, P05, P08, P10, P11, P18, P27, P38
- 13 ConnectX and BlueField: endpoints and infrastructure - C06, E01, E05, E09, P11, P19, P20, P21, P22, P24, P25, P26, P43, P48, P49, P50
- 14 Spectrum, Quantum and photonics - C06, E02, E09, E10, P23, P39, P40, P44, P47, P51, R08
- 15 Groq 3 LPX and specialized inference - E06, E09, E17, P12
- 16 Integrated systems and the bandwidth hierarchy - E08, E09, E12, E15, E16, P01, P02, P03, P06, P08, P09, P10, P11, P12, P13, P14, P16, P17, P18, P19, P20, P21, P27, P38
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